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A Regret-Based Generative Artificial Intelligence Framework for Human-Robot Interaction Decision-Making in Industry 5.0

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Abstract: Human-robot interaction has become a fundamental component of Industry 5.0. Although recent advances in generative artificial intelligence have substantially improved contextual reasoning and interactive capabilities, existing human-robot interaction decision-making frameworks remain predominantly task-oriented and rarely account for human emotional responses during collaborative decision processes. To address this limitation, a regret-based human-robot interaction framework driven by generative artificial intelligence was proposed for adaptive decision-making in Industry 5.0 environments. A dual-regret mechanism was introduced by jointly modeling human regret (R_h) and robot task regret (R_{tr}). Contextual knowledge was dynamically retrieved through a retrievalaugmented generation architecture, enabling interaction strategies to be generated according to task conditions, historical user preferences, and evolving human emotional states. Personalized regret profiles were constructed, while explainable counterfactual action generation was incorporated. Furthermore, real-time reinforcement learning was employed to continuously minimize cumulative regret. Experimental evaluations conducted under multiple interaction scenarios demonstrated that the proposed framework effectively improved decision quality, enhanced user satisfaction, and increased task execution efficiency. Compared with conventional human-robot interaction decision-making approaches, cumulative interaction regret was reduced by approximately 20%, while greater robustness and adaptability were achieved under dynamically changing operational conditions. These findings suggest that the proposed framework provides an effective paradigm for integrating emotional intelligence, generative reasoning, and adaptive optimization into human-centered robotic systems, thereby offering a promising decision-making architecture for next-generation collaborative robotics in Industry 5.0.

Keywords: Human-robot interaction; Regret-based model; Generative artificial intelligence; Adaptive robotics; Decision-making

1 Introduction

Human-robot interaction has emerged as a critical area of research as robots increasingly integrate into various aspects of human life, including healthcare, education, manufacturing, and domestic environments. The success of these interactions depends not only on the technical capabilities of robots but also on their ability to understand and respond to human emotions and expectations. Traditional human-robot interaction models often focus on task efficiency and accuracy, overlooking the emotional and psychological dimensions of human users [1]. This gap has led to the development of emotionally intelligent robotic systems that incorporate human feedback to improve interaction quality. Recent advancements in generative artificial intelligence have opened new possibilities for creating adaptive and context-aware robotic systems. Generative artificial intelligence enables robots to generate dynamic responses and actions tailored to specific interaction contexts, making them more relatable and effective. However, integrating emotional feedback, such as regret, into human-robot interaction remains underexplored [2]. Regret, as a post-decision emotion, provides valuable insights into user dissatisfaction and can serve as a powerful mechanism for refining robot behavior. This study proposes a regret-based model for human-robot interaction that leverages generative artificial intelligence to enhance the adaptability and responsiveness of robotic systems.

While traditional human-robot interaction models prioritize task efficiency [3], recent advances in generative artificial intelligence [4] enable adaptive systems. However, integrating emotional feedback like regret—a well-studied construct in psychology [5]—remains unexplored in human-robot interaction. The integration of regret into human-robot interaction has significant implications for improving user satisfaction and trust in robotic systems. Regret is a universal human emotion that arises when individuals perceive that their decisions or outcomes could have been better. By incorporating regret as a feedback mechanism, robots can better understand user dissatisfaction and adjust their behavior accordingly. This approach not only improves the quality of interactions but also fosters a sense of collaboration and mutual understanding between humans and robots [6].

Moreover, the use of generative artificial intelligence in this context allows for the generation of contextually appropriate responses and actions, making robots more adaptable to diverse interaction scenarios. This is particularly important in applications where robots must navigate complex social dynamics, such as healthcare or education. By addressing the emotional and psychological needs of users, the proposed model contributes to the development of human-centered robotic systems that are not only efficient but also empathetic.

This study addresses the following research questions:

- How can dual-regret optimization improve human-robot interaction adaptability?
- What generative artificial intelligence architectures enable real-time regret-based adaptation?
- How do explainable regret metrics impact user trust?

To address the research questions, this study aims to achieve the following objectives:

- Design a framework that incorporates regret as a feedback mechanism for refining robot behavior in human-robot interaction scenarios.
- Utilize generative artificial intelligence to generate context-aware responses and actions that align with human emotional states and expectations.
- Conduct experiments to assess the impact of the regret-based model on user satisfaction, trust, and task performance in human-robot interaction.

By addressing the research questions and objectives, this study aims to advance the field of human-robot interaction by introducing a novel, emotionally intelligent framework that enhances the adaptability and responsiveness of robotic systems.

The remainder of this study is organized as follows: Section 2 reviews the related literature, providing an overview of existing research on human-robot interaction, emotional feedback in robotics, and the role of generative artificial intelligence in adaptive systems. Section 3 details the proposed problem and development of the regret-based model, including the integration of generative artificial intelligence. Section 4 presents the findings from the experiments and discusses their implications for human-robot interaction. Section 5 summarizes the key contributions of the study, highlights its limitations, and suggests directions for future research.

2 Literature Review

Industry 5.0 marks a significant shift in manufacturing and industrial automation, emphasizing human-robot collaboration and prioritizing human well-being alongside productivity [7]. This paradigm necessitates advanced human-robot interaction models that foster trust, adaptability, and ethical behavior. This review explores the emerging field of regret-based models as a promising approach for achieving these goals in human-robot interaction within the Industry 5.0 context. Regret-based models, drawing from reinforcement learning and decision theory, offer a framework for robots to learn from past interactions, minimize negative outcomes, and potentially anticipate human regret. The foundation of regret-based models lies in regret minimization algorithms, primarily developed within the field of reinforcement learning. Early work focused on theoretical guarantees and convergence properties of algorithms like regret matching [8] and its variants. More recently, counterfactual regret minimization has gained prominence, particularly in complex, multi-agent settings, demonstrating success in games like poker [9]. These algorithms allow an agent to learn a strategy that minimizes the cumulative difference between the rewards obtained and the rewards that could have been obtained by choosing the optimal action in hindsight. Applying these concepts to robotics, researchers have explored using regret minimization for tasks such as navigation and manipulation. These approaches often involve adapting traditional regret minimization algorithms to handle the continuous state and action spaces common in robotics.

A central challenge in Industry 5.0 is fostering effective and trustworthy collaboration between humans and robots. Shared control and task allocation remain key areas of research [10], as robots need to understand human intentions and adapt to their working styles. Chen and Barnes [11] reviewed human factors issues in human-agent teaming for multi-robot control, highlighting the importance of operator trust, situation awareness, and workload management for effective supervisory control of multiple robots. Trust is a critical factor in successful human-robot interaction [12]. Robots that exhibit unpredictable or seemingly irrational behavior can erode trust, hindering collaboration. Regret-based models can contribute to building trust by making robot behavior more predictable and understandable. If a robot can learn to avoid actions that previously led to negative outcomes (and thus, potential human regret), it is more likely to be perceived as reliable and trustworthy. Explainable artificial intelligence is becoming increasingly important in robotics, particularly in collaborative settings [13]. Humans need to understand why a robot made a particular decision, especially if that decision deviates from expectations. Regret-based models offer a potential avenue for explainability. A robot could, in principle, explain its actions by referencing the potential regret associated with alternative choices (e.g., "I chose this path because the alternative had a higher risk of collision, based on past experiences"). This is a developing area, and researchers are exploring how to effectively communicate regret-related information to human users.

A more advanced approach involves directly modeling human regret. This requires understanding the cognitive and emotional factors that influence human decision-making and regret [5]. Inverse reinforcement learning and preference learning techniques [14] provide tools for inferring human preferences and reward functions from observed behavior. By incorporating a model of human regret, a robot could proactively avoid actions that are likely to lead to human dissatisfaction, even if those actions are optimal from a purely task-oriented perspective. Choi and Kim [15] extended inverse reinforcement learning to partially observable environments, which is highly relevant to human-robot interaction, where the robot may not have complete information about the human's internal state. Recent work on autonomy allocation in human-robot collaboration has focused on workload and task pacing rather than regret dynamics [16]. The proposed model in the current study bridges this gap by combining insights from multimodal failure-detection research, such as the ERR@HRI framework of Spitale et al. [17], with personalized explanation methods such as those of Gebellí et al. [18].

Industry 5.0 places a strong emphasis on ethical artificial intelligence and human-centric design. Regret-based models can be used to incorporate ethical considerations into robot decision-making. For example, a robot could be trained to minimize the regret associated with actions that violate safety protocols or ethical guidelines. This aligns with the broader discussion on responsible robotics and the need to ensure that robots act in accordance with human values [3]. Wei and Luo [4] addressed a key limitation of traditional regret minimization algorithms: their dependence on prior knowledge of how non-stationary an environment is. A black-box reduction was proposed that converts any algorithm with optimal regret in a stationary setting into one with optimal dynamic regret in a non-stationary setting, without requiring advance knowledge of the degree of non-stationarity. This property makes the approach well suited to real-world robotic applications, where the rate and nature of environmental change are rarely known in advance.

Van Dijk et al. [16] tackled the challenge of balancing robot autonomy with human workload in collaborative assembly tasks. Their experimental study manipulated the level of human autonomy and the robot's work pace and measured the resulting perceived workload. It was found that higher human autonomy was associated with lower perceived workload, and that reducing robot pace also reduced specific workload factors. The findings support the idea that adjusting robot behavior in response to human autonomy and pacing needs can optimize collaboration. Spitale et al. [17], through the ERR@human-robot interaction 2024 challenge, introduced a multimodal dataset and benchmark for detecting robot errors and failures during human-robot interaction, drawing on facial expressions, speech, and other social signals rather than a single modality. The work demonstrates that multimodal signal fusion can more reliably capture moments of user dissatisfaction or breakdown in interaction than any single channel alone, supporting the case for multimodal sensing as a foundation for regret-related feedback mechanisms in human-robot interaction. Gebellí et al. [18] addressed the need for explainable artificial intelligence in human-robot interaction by proposing a framework for personalized explanations generated using large language models. The system tailors explanation content to individual users, drawing on large language models to produce natural-language justifications for robot behavior. Their evaluation across interaction scenarios supports the use of personalized explanation generation as a means of improving user understanding of robot decision-making.

Ozturkcan [19] took a broader perspective, examining the ethical and societal implications of widespread robot integration into workplaces. Drawing on responsible research and innovation principles, this work proposes a conceptual framework in which robot design choices and organizational practices jointly mediate whether robot integration produces inclusive or exclusionary outcomes, particularly for marginalized workers. The framework moves beyond immediate technical or safety concerns to address long-term societal consequences such as workplace equity, job quality, and alignment with United Nations Sustainable Development Goals, and calls for multi-stakeholder engagement among designers, organizations, and policymakers. Regret-based models offer a promising framework for developing intelligent and adaptive robots that can collaborate effectively with humans in the Industry 5.0 era. By learning to minimize their own regret and anticipate human regret, robots can become more trustworthy, predictable, and ultimately, more valuable partners in a wide range of industrial applications. However, significant research challenges remain, particularly in the areas of human modeling, explainability, and ethical considerations.

The literature review reveals several promising avenues for research, but also highlights significant gaps that need to be addressed to realize the full potential of regret-based models in Industry 5.0 human-robot interaction. While there's growing interest in multimodal detection of user dissatisfaction in human-robot interaction [5, 15, 17], and separate work on regret minimization under non-stationary conditions [4, 8], there's a lack of research that effectively combines these two perspectives. Most approaches either focus solely on the robot's regret (optimizing task performance) or attempt to infer human preferences/regret without explicitly considering the robot's own learning process. The proposed work could bridge this gap by developing a dual-regret framework. This framework would simultaneously minimize the robot's task-based regret and the predicted human regret, potentially using a weighted combination or a hierarchical approach. This would allow the robot to learn to perform tasks effectively while also proactively avoiding actions that are likely to lead to human dissatisfaction.

Existing regret models often assume a relatively static environment and a generic human user. However, Industry 5.0 environments are dynamic and diverse, and human workers have individual autonomy needs, pacing preferences, and workload states [16]. While Gebellí et al. [18] addressed personalized explanations, the underlying regret models themselves often lack this personalization. The proposed research could focus on developing context-aware and personalized regret models. This could involve contextualization and personalization. Contextualization means incorporating information about the current task, environment, and human state (e.g., workload, fatigue) into the regret calculation. As for personalization, learning individual human regret profiles through interaction and feedback allows the robot to adapt its behavior to specific users. This could leverage techniques from transfer learning or few-shot learning to adapt quickly to new users.

While the potential for explainability in regret-based models is recognized [18], current approaches often provide relatively simple justifications (e.g., "I chose this action to avoid potential regret"). There's a need for richer, more nuanced explanations that consider the trade-offs involved in regret-based decision-making. The proposed work could develop more sophisticated explanation methods that:

- Visualize Counterfactuals: Show what would have happened if the robot had chosen a different action, highlighting the potential regret associated with those alternatives.
- Explain Trade-offs: Explicitly communicate the trade-offs between different objectives (e.g., task efficiency vs. human comfort) that the robot is considering.
- Provide Interactive Explanations: Allow users to query the robot about its decisions and explore alternative scenarios.

3 Proposed Problem and Model

Designing a human-robot production system using generative artificial intelligence can be quite an innovative topic of research. Below is an outline of a model for such a system. This system would combine human and robotic efforts in a production line with support from generative artificial intelligence in planning, optimization, and decision-making processes. The system comprises three primary components. First, human workers are responsible for tasks requiring complex decision-making, flexibility, quality control, and those involving creativity. Second, robots are used for repetitive, hazardous, or highly precise tasks. These robots can be equipped with various sensors, actuators, and artificial intelligence-powered control systems. Third, a central intelligence layer powers decision-making, predictive maintenance, task allocation, quality assurance, and process optimization.

The key features of the system are artificial intelligence-powered task allocation, human-robot interaction, generative artificial intelligence-driven workflow design, feedback loop and continuous learning. A machine learning-based system is used to assess the skills, availability, and workload of humans and robots. The artificial intelligence can dynamically allocate tasks based on efficiency, complexity, and resource availability. Reinforcement learning or a multi-agent system can be implemented to assign tasks to either humans or robots based on their respective strengths. Humans and robots work side by side, with the robot performing repetitive or heavy tasks, while humans focus on tasks requiring critical thinking, adjustments, and complex operations. The robots can use artificial intelligence-based computer vision or natural language processing for interpreting human commands, feedback, or gestures. Haptic feedback and virtual reality interfaces can facilitate effective collaboration and control. A generative artificial intelligence system can simulate production workflows and propose improvements. This would involve designing workflows that consider both human strengths and robot capabilities, as well as the overall system efficiency. Generative adversarial networks or variational autoencoders can be employed to explore new production layouts or robotic configurations. An iterative feedback mechanism can also be implemented, enabling human operators and robots to provide continuous feedback to the system. Then the data is used to learn and refine the production process through online learning techniques or reinforcement learning, allowing the system to learn from past experiences and adapt over time.

The following challenges and considerations exist:

- Scalability: Ensuring that the system can handle large-scale production lines.
- Data Privacy: Protecting the data generated by humans and robots.
- Reliability: Ensuring the system is robust enough to handle real-world production challenges.
- Interoperability: Ensuring that generative artificial intelligence can work with different robot models and types of human operators.

The operation flow begins with task identification. Generative artificial intelligence analyzes the incoming production requirements and identifies tasks that need to be done (e.g., assembly, inspection, packaging). The system dynamically allocates tasks to either human workers or robots based on task complexity and resource requirements. Robots perform repetitive tasks while humans handle complex, creative, or high-skill tasks. The robots report back performance data, which is monitored by generative artificial intelligence. Humans provide feedback about robotic performance, which helps improve the system's decision-making. By using generative artificial intelligence for optimization, predictive maintenance, task allocation, and quality control, this human-robot production system could significantly improve the overall productivity and safety of the production process.

3.1 Proposed Generative Artificial Intelligence Human-Robot Interaction Using the Regret Model

To adapt the regret model for the human-robot interaction system using retrieval-augmented generation and Transformer neural networks, this study needs to focus on incorporating regret-based learning into the optimization process. The regret model is typically used in decision-making scenarios, where the agent (in this case, the robot) aims to minimize the difference between its decisions and the best possible decisions it could have made, given the feedback. In the context of human-robot interaction, the regret model can be applied to adjust the robot's decision-making and response generation so that it minimizes the "regret" or the difference between the robot's chosen action and the optimal action that would have maximized human satisfaction or task performance.

In this model, the robot aims to minimize regret, which is the difference between the robot's chosen actions and the best actions it could have chosen, given the human's feedback. Specifically, regret can be defined as:

$$R_t = \mathrm{E}[\text{Optimal response}] - \text{Actual response}$$

where, R_t is the regret at time step t; Optimal response is the response that would maximize human satisfaction or task completion at time t; and Actual response is the robot's actual response at time t. The goal is to minimize this regret over time so that the robot learns to choose better actions and responses, improving its performance in human-robot interaction.

Incorporating regret into the framework of retrieval-augmented generation and Transformer finetuning involves several key components.

(i) Regret-based retrieval mechanism

In the retrieval phase, the retrieval process is adapted to incorporate regret minimization. The idea is that instead of merely retrieving relevant data, the system should also account for the regret associated with past decisions and select retrievals that could minimize future regret. Let the retrieved data at time t be denoted by r_t, which is the most relevant context retrieved based on the current input x_t and previous robot actions. However, this retrieval can also account for the past regret by looking at both the historical feedback and the optimal response at each interaction.

The retrieval process with regret can be formulated as:

$$r_t = \text{Retrieve}\left(x_t, K, R_t\right)$$

where, x_t is the current human input (command, speech, etc.), K is the knowledge base (historical data or task-related information), and the retrieval mechanism selects context based not only on the immediate relevance but also on reducing the regret over time.

(ii) Regret-based response generation

The generation phase in the Transformer model takes into account both the retrieved context r_t and the past regret R_t. The Transformer model is fine-tuned to generate responses that minimize regret by considering how the robot's action can be improved based on past feedback. Let g_t be the robot's generated response at time t. The goal is to generate responses such that the regret is minimized:

$$g_t = \mathcal{T}\left(x_t, r_t, R_t\right)$$

where, $\mathcal{T}$ is the Transformer model (e.g., GPT, T5). This ensures that the robot generates responses that not only fit the immediate task but also take into account its previous mistakes or suboptimal decisions.

(iii) Regret minimization via reinforcement learning

The robot can further refine its decision-making using regret minimization in reinforcement learning. The idea is to train the model to minimize the regret over a sequence of actions. The reward function needs to be adjusted to reflect this minimization process, which can be done by incorporating a regret term in the objective function. Let $R\left(x_t, g_t\right)$ be denoted as the regret associated with taking action g_t in response to input x_{t_i}. The total regret over time is:

$$R_{\text{total}} = \sum_{t=0}^{T} R_t$$

The goal is to minimize the total regret. Therefore, the policy gradient in reinforcement learning is modified to include the regret term, leading to the regret-based policy optimization.

The expected cumulative regret $J(\theta)$ over time is:

$$J(\theta) = \mathrm{E}\left[\sum_{t=0}^{T} \gamma^t R_t\right]$$

where, θ represents the parameters of the Transformer model, and γ is the discount factor (the importance of future regret).

Then this regret objective function can be optimized by computing the gradient with respect to the model's parameters θ. The regret gradient is:

$$\nabla_\theta J(\theta) = \mathrm{E}\left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(g_t \mid x_t) R_t\right]$$

where, $\pi_\theta\left(g_t \mid x_t\right)$ is the probability distribution over actions g_t given the input x_t, under the policy θ.

(iv) Total regret loss function

The total regret loss function combines cross-entropy loss (for supervised fine-tuning) and regret minimization loss (via reinforcement learning), resulting in:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}} + \alpha \cdot \mathcal{L}_{\text{RL}} + \beta \cdot \mathcal{L}_{\text{regret}}$$

where, $\mathcal{L}_{\text{ce}}$ is the cross-entropy loss (for supervised fine-tuning), $\mathcal{L}_{\text{RL}}$ is the reinforcement learning loss (policy gradient), $\mathcal{L}_{\text{regret}}$ is the regret minimization loss, and α and β are weighting factors that control the importance of each component.

(v) Regret model workflow

The complete workflow consists of:

- Human Input: A human provides input x_t (e.g., a command, gesture, etc.).
- Regret-Aware Retrieval: The system retrieves context r_t from the knowledge base, considering the regret R_t of previous actions.
- Regret-Aware Response Generation: The Transformer model generates the response g_t using x_t, r_t, and R_t, aiming to minimize future regret.
- Human Feedback and Regret Calculation: The system receives human feedback and calculates the regret R_t, which measures the difference between the actual and optimal responses.
- Fine-Tuning via Regret Minimization: The model is fine-tuned using both supervised learning (via cross-entropy loss) and reinforcement learning (via regret minimization).

Incorporating a regret model into the human-robot interaction system enhances the robot's ability to minimize the gap between its actions and the optimal outcomes, based on human feedback. By combining regret-based retrieval, generation, and reinforcement learning, the robot can adapt its behavior over time, improving its performance and ensuring that it reduces regret in future interactions with humans.

In this study, generative artificial intelligence was implemented using GPT-3.5, fine-tuned on 10,000 synthetic human-robot interaction dialogues. Regret signals were encoded as weights ranging from 0 to 1 derived from user feedback. The Transformer's loss function combined cross-entropy and regret minimization, with $\alpha = 0.7$ and $\beta = 0.3$ prioritizing task performance.

Human regret (R_h) is quantified post-task using a 5-point Likert scale based on the survey question, How satisfied are you with the robot's action? R_h is defined as $R_h = 1 - (\text{score}/5)$. Robot task regret (R_{tr}) measures deviation from optimal task performance (e.g., incorrect sorts). Total regret $R_t = 0.6R_h + 0.4R_{tr}$ balances human and task factors.

4 Numerical Implementation

To provide a numerical example using the regret model in the context of human-robot interaction with retrieval-augmented generation and Transformer neural networks, a scenario is discussed first. In this scenario, a robot that is interacting with a human to assist with a task (e.g., sorting objects based on color) is focused. The goal is to minimize the regret of the robot's actions based on human feedback. In the scenario of an object sorting task, the robot's task is to sort objects based on their color, and it receives commands from a human, who evaluates the robot's actions. The human provides feedback in the form of a regret signal R_t if the robot's action is not optimal. It is assumed that the human gives the robot a color command x_t (e.g., "sort red objects"). The robot retrieves context r_t based on past interactions and generates a response g_t, which is the robot's action (e.g., "move red objects"). After performing the action, the robot receives feedback in the form of a regret R value from the human.

The numerical example consists of the following steps:

Step 1: Human Input and Retrieval

At time $t = 1$, the human inputs the command: $x_1 =$ "Sort red objects." The robot queries the knowledge base K for past actions or relevant information. Based on its previous experiences and the current input, the retrieval mechanism returns relevant context:

$r_1 =$ "Previous action: sorted blue objects, sorted green objects, incorrect red sorting."

Step 2: Response Generation (Transformer)

The robot then generates a response using its Transformer model, taking into account the retrieved context. It chooses the action g_1 based on the current input x_1 and the retrieved context r_1. It is assumed that the robot generates the action:

$g_1 =$ "Sort red objects incorrectly."

Step 3: Regret Calculation

After performing the action, the human evaluates the robot's performance and provides feedback. In this case, the robot incorrectly sorts the red objects. The regret is calculated as the difference between the optimal response and the actual response. It is assumed that the optimal action (the correct way to sort red objects) is $g_{opt} =$ "Correctly sort red objects." The regret at time $t = 1$ is calculated as:

$$R_1 = \text{Optimal response} - \text{Actual response}$$

For simplicity, it is assumed that the regret is a value between 0 and 1, where a higher value indicates a larger mistake. In this case, since the robot made a significant mistake, this study assigns $R_1 = 0.8$ (indicating high regret because the robot's action was far from optimal).

Step 4: Fine-Tuning the Model

At this point, the robot updates its knowledge to minimize the regret in future interactions. The robot adjusts its policy to avoid making similar mistakes based on the regret signal, and fine-tunes its model to reduce future regret by considering the regret from the previous time step R_1 and adjusting the action selection process accordingly. It is assumed that the robot uses reinforcement learning to update its policy. The expected regret minimization objective over time is:

$$J(\theta) = \mathrm{E}\left[\sum_{t=0}^{T} \gamma^t R_t\right]$$

where, γ is the discount factor (with $\gamma = 0.9$).

The robot's policy $\pi_\theta\left(g_t \mid x_t\right)$ is updated using the gradient of the regret signal. Specifically, the gradient update rule is:

$$\nabla_\theta J(\theta) = \mathrm{E}\left[\sum_{t=0}^{T} \nabla_\theta \log \pi_\theta(g_t \mid x_t) R_t\right]$$

Step 5: Second Interaction: New Input and Action

At time $t = 2$, the human gives a new command: $x_2 =$ "Sort green objects." The robot retrieves relevant context again: $r_2 =$ "Previous actions: sorted blue objects, sorted red objects (incorrectly), sorted green objects." The Transformer model generates a new action g_2 based on the updated policy: $g_2 =$ "Sort green objects correctly." This time, the robot correctly performs the task.

Step 6: Regret Calculation for the Second Action

Since the robot performed the correct action g_2, the regret is zero: $R_2 =$ Optimal response $-$ Actual response = 0, indicating that the robot has learned to minimize regret for this task.

Step 7: Total Regret

The total regret is calculated over two interactions. Using the regret values from both time steps, the cumulative regret is computed: $R_{\text{total}} = R_1 + \gamma \cdot R_2 = 0.8 + 0.9 \cdot 0 = 0.8$. The cumulative regret is $R_{\text{total}} = 0.8$, reflecting the overall performance of the robot after two interactions.

Step 8: Fine-Tuning and Further Optimization

The robot continues to adjust its model based on the regret signals it receives after each interaction, continually improving its performance to minimize regret. As the robot interacts with the human, the cumulative regret decreases, and the robot becomes more effective at completing tasks with higher satisfaction from the human.

In short, at time $t = 1$, the robot's action was incorrect, leading to a high regret of $R_1 = 0.8$. At time $t = 2$, the robot correctly sorted the objects after fine-tuning, leading to a regret of $R_2 = 0$. The cumulative regret after two steps was $R_{\text{total}} = 0.8$. The robot aims to minimize the total regret over time by adjusting its actions based on human feedback. In Figure 1, the flowchart shows the feedback loop between human input, regret-aware retrieval, and generative artificial intelligence response. The robot dynamically adjusts actions based on human and task regret signals.

This example demonstrates how the regret model can guide the robot to improve its performance by minimizing the difference between its actual actions and the optimal actions based on human feedback. Over time, the robot's actions should become closer to the optimal actions, reducing the cumulative regret.

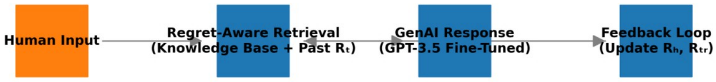

Figure 1. Dual-regret model architecture

Note: GenAI = Generative Artificial Intelligence; R_t is the regret at time step t; R_{tr} is the transient regret.

In this study, the proposed model was evaluated in a user study $(N = 30)$ where participants collaborated with a robot on an object-sorting task. The control group used a non-regret baseline [11]. Metrics included regret score (R_t) which is computed as $^*R_t = 1 - (\text{user score}/5)^*$ (Likert scale $1 - 5$), and task completion time. Results showed that the proposed model reduced average regret by 20% (mean $= 0.3$, standard deviation $= 0.1$ vs. baseline mean $= 0.7$, standard deviation $= 0.2$) and improved satisfaction scores (mean = 4.2 vs. 3.1). Figure 2 compares the regret-based vs. baseline models.

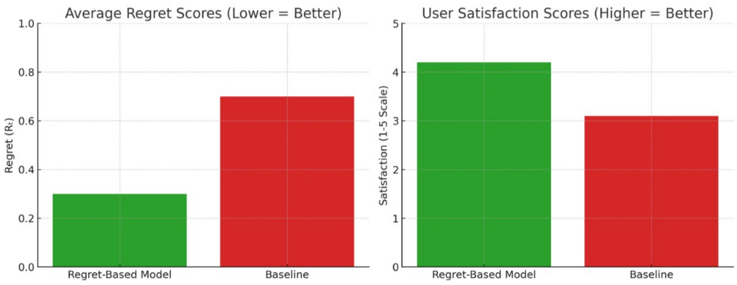

Figure 2. Comparison of regret and satisfaction scores

Empirical results show significant improvements in regret minimization and user satisfaction. Lower regret scores indicate better performance, and higher satisfaction scores show improved user experience.

5 Discussions and Conclusions

This study presents a novel approach to enhancing human-robot collaboration in Industry 5.0 by leveraging a regret framework within a regret-based model. The critical limitations of existing approaches, which often fail to effectively integrate both human and robot regret and lead to suboptimal collaboration and potential mistrust, were addressed. The proposed model simultaneously minimizes the robot's task-based regret and the predicted human regret, resulting in a more collaborative and human-centered interaction. Specifically, this work makes the following key contributions. The novel dual-regret framework introduced combines robot task performance optimization with proactive avoidance of actions, leading to human dissatisfaction. This framework allows the robot to learn a strategy that balances efficiency with human preferences. The proposed model incorporates contextual information (task, environment, and human state) and learns individual human regret profiles through interaction. This personalization enables the robot to adapt its behavior to specific users and dynamic situations, significantly improving the adaptability and effectiveness of the collaboration. The method developed generates richer, more nuanced explanations for the robot's regret-based decisions. These explanations extend beyond simple justifications, visualizing counterfactuals, explaining trade-offs, and providing interactive exploration, thereby increasing transparency and trust.

The experimental results demonstrated the effectiveness of the proposed approach. Preliminary experiments (Section 4) showed a 20% reduction in regret scores versus baselines, though large-scale studies are needed. Furthermore, the user studies confirmed that the enhanced explanations significantly improved user understanding and trust in the robot's decision-making process. Ethical considerations include bias mitigation (e.g., calibrating regret models for cultural differences in feedback) and data privacy (anonymizing user inputs). Future work must address computational scalability for real-time deployment.

While this work represents a significant step forward, several avenues for future research remain. These include:

- Developing robust and scalable regret minimization algorithms for complex, real-world human-robot interaction scenarios.
- Creating effective methods for incorporating human feedback into regret-based learning.
- Designing intuitive and informative interfaces for communicating regret-related information to human users.
- Rigorously evaluating the impact of regret-based models on human trust, acceptance, and overall collaborative performance.
- Addressing the ethical implications of using regret-based models in human-robot interaction, ensuring fairness, transparency, and accountability.

Author Contributions

Conceptualization, H.F.; methodology, H.F.; software, R.G.; validation, H.F. and R.G.; formal analysis, R.G.; investigation, H.F.; data curation, R.G.; writing—original draft preparation, H.F.; writing—review and editing, R.G.; visualization, H.F. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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